AWS and EC2 Instances

What is security group

1. it is a virtual firewall works at ENI (instance) level.
2. up to 5 security groups per EC2 instance interface can be applied.
3. can only have permit rules, cannot have deny rule.
4. And its stateful means the return traffic is automatically allowed
5. if you allow port in inbound traffic then you cannot attach separate port in outbound traffic the port will be automatically allow both inbound and outbound traffic.

Base points

1. it is operated at instance level.
2. support allows rules only.
3. stateful, return traffic is automatically allowed.
4. applies to an instance only

Inbound (Incoming):
1. Traffic coming into your server
2. Controls who can access your instance

Outbound (Outgoing):
1. Traffic going out from your server
2. Controls where your instance can connect

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Create a Instance Windows & get the access

1. create a instance

1.name----------> Windows_Server-2016-English-Full-Base-2026.02.11
2.instance type------> t3. micro
3.security group----> ssh / http / RDP (3389)
4.create a instance

2.access the instance

1.connect the instance
2.go to RDP client
3.click on get password------> browse key pair-----> decrypt password ---> copy
4.open RDP in desktop
5.computer---> (public IP)

6.username---> (Administrator)
7.password----> (paste password)
8.connect

3.Install webserver and create a webpage on windows server

1.search server manager and click
2.add roles and features
3.next
4.role based or features based installation
5.next
6.select a server from server pool
7.next
8.select one file------> webserver IIS------add features-------next------next-----next-----install-----> close
9.click on file manager
10.click on c drive
11.inetpub-----> wwwroot ------> 2files (delete these files)
12.create a index.html file but open the notepad ---> save as index.html ---> all files "HELLO WORLD"

4.(public IP) ---------->check the web

Create a Instance - amazon Linux & get the access

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1.create a instance

1. name ----------> amazon Linux
2. instance type --> t3. micro
3. key name -------> amazon-linux
4. vpc ----------> default
5. sg ----------> ssh(22) / http(80)
6. create a instance

2.Get access of instance

1. open the powershell terminal on the windows
2. Run the command
    1. chmod 600 amazon-linux.pem
    2. ssh -i “<key_name>” ec2-user@<public_ip>
       ssh -i “amazon-linux.pem” ec2-user@<public_ip>

Note:

- ssh → connect command
- -i → identity file (key pair)
- amazon-linux.pem → your private key file
- ec2-user → default user for Amazon Linux
- <Public-IP> → instance public IP

What is private ip and public ip

1.Public ip

1. Public IP is assigned by the ISP (internet service provider)
2. It is a globally unique (no other devices on the internet having the same public IP).
3. It is routable over the internet
4. It is a paid

2.Private ip

1. Private IP is Assigned by system admin / router / DHCP inside the LAN.
2. Locally unique only (can be repeated in different networks).
3. Not routable over the internet directly (requires NAT to communicate outside).
4. Free of cost (anyone can use private IP ranges).

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